Type 2 Paper: A Critical Methodological Review AI Governance in Public Sector and Implications

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Introduction

The first paper authored by Kuziemski and Misuraca, investigates the deployment of Artificial Intelligence (AI) in the public sector through three case studies: Canada, Poland, and Finland. It explores how AI is influencing governance, focusing on legal frameworks, implementation challenges, and governance practices. The authors aim to provide insights into the implications of automated decision-making (ADM) in democratic societies, emphasizing the tension between enhancing efficiency and maintaining accountability, transparency, and fairness. The paper critically examines the power asymmetries created by AI in governance, highlighting the need for a unified framework to evaluate AI's impacts on public administration. It analyzes the goals, drivers, barriers, and risks associated with AI deployment, it offers a comparative perspective on three nations with diverse socio-economic models and regulatory approaches. On the other hand, the second paper titled "Implications of the Use of Artificial Intelligence in Public Governance" by Zuiderwijk, Chen, and Salem systematically examines the transformative potential and associated challenges of AI applications in the public sector. By analyzing 26 scholarly articles, the paper addresses the increasing complexities surrounding AI’s integration into government operations, particularly its implications for governance, decision-making, and public policy. It sets the stage for a broader research agenda aimed at addressing current knowledge gaps and proposing effective strategies for AI deployment in public governance.

Keywords

AI Governance, Automated Decision-Making (ADM), Public Administration, Ethical AI Frameworks, Transparency and Accountability, Systematic Literature Review, Data-Driven Decision-Making, Governance Gaps, Algorithmic Bias. Transformer Models

Summary of Key Findings

The study underscores the big potential of AI in governance, focusing on its application in diverse contexts across the three countries. In Canada, AI systems are primarily used to automate immigration processes. The key goals include improving efficiency, reducing backlogs, and enhancing transparency through tools like the Algorithmic Impact

Assessment (AIA). Canada’s strong research infrastructure and funding initiatives, such as the Pan-Canadian AI Strategy, drive these efforts. However, challenges such as algorithmic opacity, ethical dilemmas, and public mistrust regarding automated systems in high-stakes scenarios persist. Risks include potential biases in algorithmic decision-making and limited recourse for individuals affected by adverse outcomes.

In Poland, AI has been adopted to optimize public employment services by automating the profiling of job seekers. The primary goal is to allocate limited resources more effectively, driven by the need to modernize public services and address structural unemployment. However, the reliance on opaque algorithms and the lack of transparency have led to public backlash and legal challenges, with the Constitutional Tribunal ruling parts of the profiling system unconstitutional. Risks associated with this approach include the marginalization of vulnerable groups and the erosion of public trust in government institutions. Finland’s approach to AI governance emphasizes ethical and human-centric applications. Finland also integrates AI into public service delivery, tailoring services to individual citizens' needs. The initiative is driven by Finland’s strong data governance practices, such as MyData, and its commitment to inclusivity and well-being. While Finland’s model highlights the potential for equitable and citizen-focused AI applications, barriers such as cultural resistance to change, resource constraints, and the complexity of cross-departmental collaboration pose challenges. Risks include the need for robust data protection measures and legal frameworks to ensure the ethical use of AI.

The second paper identifies a range of opportunities and risks associated with AI in public governance. Among the opportunities, AI enhances efficiency by automating repetitive tasks, simplifying processes, and reducing errors. It supports data-driven decision-making by providing more accurate insights that inform policies. Personalized public services are another significant benefit, where AI enables governments to tailor service delivery to individual citizen needs, thereby increasing accessibility and satisfaction. Moreover, AI contributes to risk management by assisting in fraud detection, urban safety monitoring, and disaster prediction.

Despite these benefits, the paper also highlights substantial challenges. Ethical concerns, such as data privacy, algorithmic bias, and a lack of transparency in AI systems, pose

significant risks. These issues undermine fairness and trust in public administration. Governance gaps, caused by the rapid pace of AI development outstripping regulatory responses, exacerbate these challenges. Technical barriers such as poor data quality, integration issues, and insufficient expertise further hinder effective AI implementation.

To address these complexities, the authors propose a detailed research agenda. Process-related recommendations emphasize the transition from exploratory to empirical research, fostering multidisciplinary collaborations, and focusing on practical applications of AI. Content-related priorities include developing theoretical governance frameworks, investigating stakeholder engagement strategies, and assessing AI's societal impacts. This agenda aims to fill existing knowledge gaps and guide the responsible deployment of AI in public governance.

Comparative Methodology and Discussion

The methodology employed in this study is a comprehensive qualitative approach designed to explore the implications of AI governance in the public sector across three distinct democratic settings: Canada, Poland, and Finland. The analysis is framed around four thematic areas—goals, drivers, barriers, and risks—which provide a structured yet comprehensive lens to evaluate the adoption of AI in public services. The study begins with a detailed literature review to establish a foundational understanding of AI governance. By synthesizing regulatory frameworks, policy documents, and academic research, the literature review contextualizes the case studies within broader global trends, highlighting the intersection of technical, ethical, and socio-political dimensions. It is based on semi-structured interviews conducted with a diverse group of stakeholders, including public servants, delivery leads, academics, and government contractors. This flexible interview format encouraged participants to share in-depth perspectives while addressing specific themes such as implementation challenges, ethical considerations, and performance metrics. The insights from these interviews added valuable real-world dimensions to the study, revealing how AI systems are perceived, implemented, and governed.

The methodology also incorporates case study analysis, selecting Canada, Poland, and Finland based on their strong Government AI Readiness Index rankings and diverse socio-economic contexts. Each case study focuses on a distinct application of AI in public services: Canada's automated immigration processes, Poland's employment service optimization through profiling, and Finland's citizen-centric AuroraAI program. The analysis within each case is organized around the thematic areas of goals, drivers, barriers, and risks, ensuring a consistent framework for comparing and synthesizing findings.

This multi-method approach contextualizes each case within its unique socio-economic and political setting while enabling a comparative analysis of shared challenges, best practices, and lessons. By clustering findings into thematic categories, the study highlights key trade-offs and implications of AI adoption, offering actionable insights for policymakers and stakeholders seeking to navigate the complexities of AI governance. The forward-looking perspectives incorporated into the analysis further strengthen its relevance, providing pathways for enhancing government legitimacy, fostering citizen trust, and achieving sustainable development through responsible AI integration. This methodology ensures a rich, nuanced understanding of the intersection between AI and public governance.

Canada: Automating the Immigration Process

Canada's implementation of AI in public governance centers around automating immigration processes. The system employs predictive tools to streamline the review of immigration applications, identifying potential red flags and recommending acceptance or rejection. The effort aligns with the Pan-Canadian AI Strategy, which emphasizes innovation, transparency, and minimizing algorithmic risks.

Goals

The primary goal is to enhance efficiency in processing immigration applications, addressing the growing volume of applicants. The automated decision-making system aims to reduce manual errors and improve service delivery while adhering to ethical standards such as fairness and transparency.

Drivers

Key drivers include the increasing demand for immigration services, resource constraints, and the ambition to position Canada as a leader in responsible AI deployment. The government has introduced measures such as the Directive on Automated Decision-Making to ensure compliance with ethical principles.

Barriers and Risks

Challenges include public skepticism, algorithmic biases, and concerns about the opacity of decision-making processes. Moreover, the potential for systemic inequalities and the difficulty of appealing algorithmic decisions exacerbate risks.

Poland: Optimizing Employment Services

Poland's use of AI focuses on optimizing employment services through automated profiling mechanisms. The system categorizes unemployed individuals into distinct profiles based on various socio-economic factors, determining the level of support they receive. This approach aims to enhance the allocation of limited resources while improving labor market outcomes.

Goals

The initiative seeks to address inefficiencies in public employment services by implementing a data-driven approach to categorize and prioritize job-seekers based on their employability and needs.

Drivers

The economic impact of the 2008 financial crisis and structural unemployment issues motivated the adoption of this system. Poland's broader industrial strategy, emphasizing technological innovation, further propelled the initiative.

Barriers and Risks

The opaque nature of the profiling algorithm and its potential for discriminatory outcomes have raised significant concerns. Legal challenges and public backlash culminated in the Constitutional Tribunal ruling the profiling tool unconstitutional.

Finland: Personalized Public Services through AuroraAI

Finland's AuroraAI program exemplifies a human-centric approach to AI governance, focusing on personalized service delivery. The platform leverages AI to assist citizens in accessing timely and relevant public services, tailored to their life events.

Goals

AuroraAI aims to revolutionize public administration by prioritizing citizen needs and fostering collaboration between various service providers. The initiative reflects Finland's commitment to well-being and inclusivity.

Drivers

The program aligns with Finland's broader strategy to enhance digital skills and data-driven governance. Demographic challenges, such as an aging population, have also driven the adoption of AI to improve service efficiency.

Barriers and Risks

Challenges include the complexity of implementing a cohesive legal framework and ensuring robust data protection mechanisms. Stakeholders have expressed concerns about the program's scalability and the cultural shift required for its success.

The second study employs a systematic literature review (SLR) methodology, following a structured and replicable process based on Kitchenham’s framework. The process begins with study identification, where the authors search three primary databases—Web of Science, Scopus, and the Digital Government Reference Library (DGRL)—to locate relevant literature. Complementary searches on Google Scholar ensure comprehensive coverage. Keywords such as "artificial intelligence," "machine learning," "public governance," and "digital government" guide the search process.

This study selection is then conducted, focusing on peer-reviewed journal articles and conference proceedings published between 2010 and 2020. The inclusion criteria ensure that selected studies place AI at the center of their investigation and relate directly to public governance. From an initial pool of 137 articles, the authors narrow their selection to 26 studies through a rigorous process of relevance and quality assessment.

Relevance is determined by evaluating whether AI and governance are central themes in each study. Quality assessment considers factors such as the clarity of research objectives, the robustness of methodologies, and the depth of data analysis. Data extraction follows, systematically recording metadata such as publication details, research objectives, methods, and findings in a structured format. This step ensures consistency and thoroughness in analyzing the selected studies.

Finally, data synthesis integrates findings from the reviewed articles to identify recurring themes and patterns. The authors categorize these themes into opportunities, challenges, and implications for public governance. This structured synthesis forms the basis for the research agenda proposed in the paper.

Conclusion

The two papers collectively explain the transformative potential and inherent complexities of integrating AI into public administration. Together, they provide a comprehensive perspective on the opportunities, challenges, and implications of AI adoption across diverse democratic settings, while emphasizing the need for robust governance frameworks to ensure equitable and effective outcomes.

Three case studies in Canada, Poland, and Finland explore AI's influence on governance practices, regulatory approaches, and decision-making processes. It highlights how these countries employ AI to address distinct goals, such as automating immigration processes, optimizing employment services, and personalizing citizen-centric public services. The analysis reveals that, while AI enables efficiency and innovation, significant barriers like algorithmic opacity, ethical concerns, and governance gaps persist. The second paper takes a broader approach, systematically reviewing 26 scholarly articles to assess the implications of AI in public governance. It identifies key benefits, such as enhanced efficiency, data-driven decision-making, and personalized service delivery, while also pointing out substantial challenges like ethical risks, data privacy issues, and the rapid pace of AI development outstripping regulatory capabilities.

Both papers emphasize common themes in AI governance, including the importance of transparency, accountability, and ethical considerations. In Canada, tools like the

Algorithmic Impact Assessment (AIA) demonstrate the potential for embedding ethical frameworks into AI applications, while Finland's AuroraAI program highlights a human-centric approach prioritizing inclusivity and well-being. Conversely, Poland's reliance on profiling algorithms underscores the risks of algorithmic biases and inadequate public engagement. The comparative methodology used in the first paper enriches our understanding of context-specific applications, while the literature review methodology in the second paper provides a broad-based analysis of governance trends and research priorities.

Conclusion

This study provides a comprehensive overview of the implications of AI in public governance, highlighting both its transformative potential and associated risks. The systematic literature review methodology ensures a robust and replicable analysis, while the proposed research agenda offers actionable insights for scholars and policymakers. As AI continues to reshape public administration, the findings underscore the importance of balancing innovation with ethical and governance considerations. The paper serves as a foundational resource for navigating the complexities of AI integration in the public sector.

References

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